AUS racing

NEWSLETTER

JULY 2024

DESIGNING A BATTERY

STARTING FROM SMALL STEPS

CREATING A CHASSIS

WHAT IT TAKES TO MAKE A GOOD CHASSIS

SPONSOR US TODAY!

LEARN HOW TO SUPPORT THIS PROJECT

WHAT CAN YOU EXPECT?

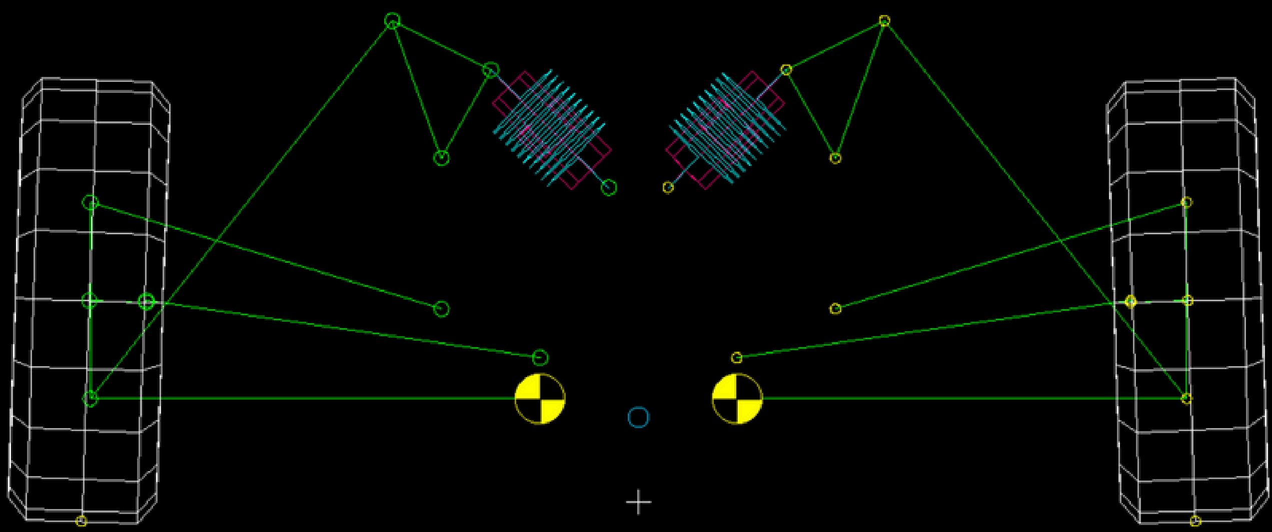
WHAT IS AUS RACING

AUS Racing is the Formula Student team from the American University of Sharjah. Formula Student is an international engineering design competition that graciously provides a platform to aspiring engineers to build an open-wheel, single seater racecar and simulate all the different nuances that culminate to account for its structural integrity, performance, economic feasibility and industrial standards.

Students compete in different static and dynamic events to assess the car, thus testing the engineering design, management skills and the overall team spirit of the contesting teams. It is an esteemed competition that unlocks new horizons for AUSers to venture into as we progress into a new era of technological competence and innovation.

Here at AUS Racing, we hope to open up opportunities for students for the foreseeable future to take part in a project that will teach them engineering, management and budgeting. In essence, real world engineering.

GETTING MORE GRIP



The suspension's main job is to ensure as much of the tire is in contact with the ground and is producing the ideal cornering when needed.

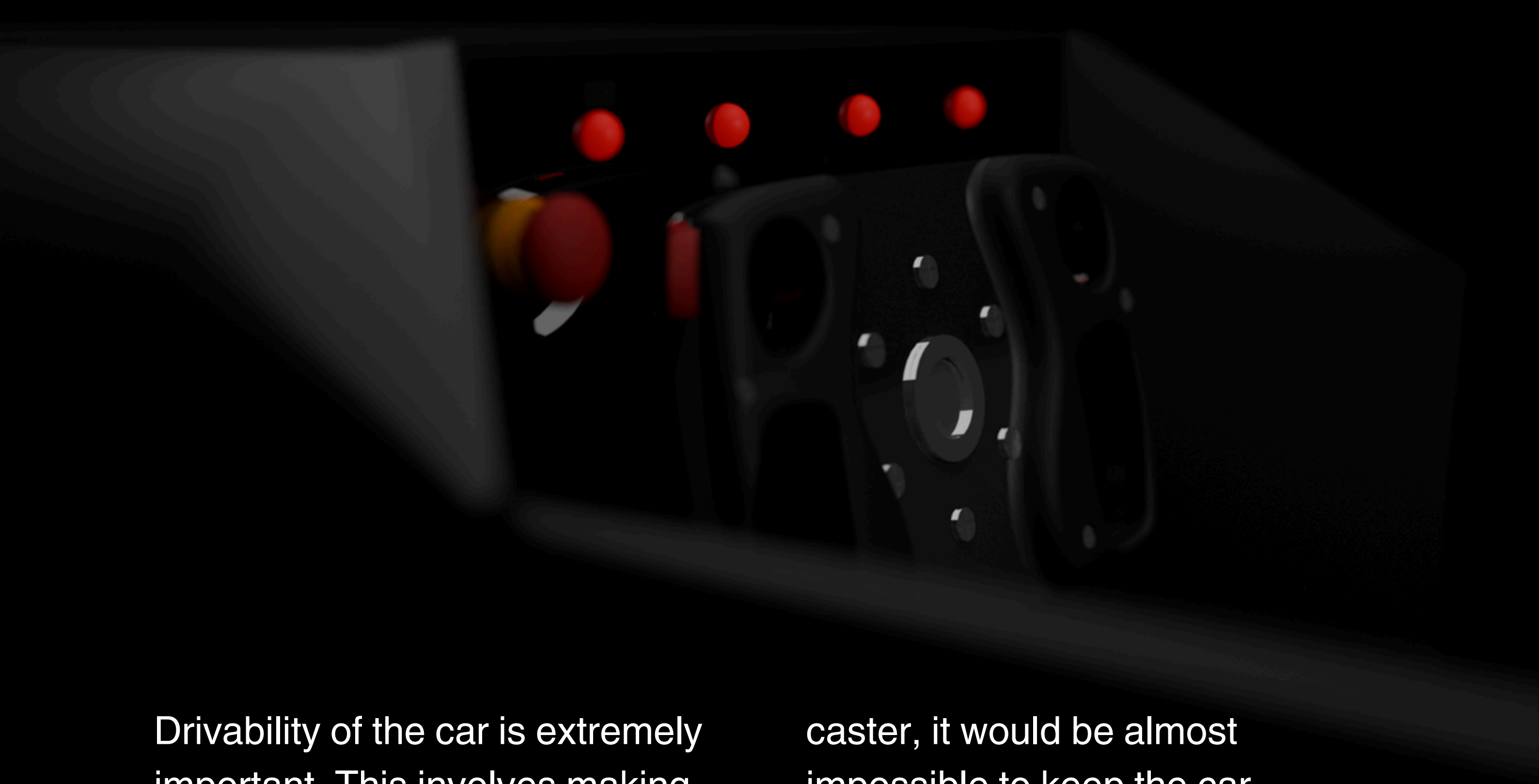
However, by relying completely on geometry, achieving these traits is not possible. Therefore, there are mechanisms in place to isolate the influence of the suspension geometry from the vehicle dynamics. By taking advantage of this we can mitigate other issues introduced by the suspension geometry.

Without such systems, the point around which the car rolls, called the roll center, would have to be extremely high. This causes jacking, an upward force created when the car rolls.

If the height of the roll center is reduced too much, the car will generate a downward force and can geometrically make the tires scrub and cause tire wear.

We found the sweet spot in the geometry where there is little to no jacking or scrub.

MANUAL STEERING



Drivability of the car is extremely important. This involves making the steering both stable and easy to use.

Have you ever wondered why your car becomes difficult to reverse at high speeds? This is because of “caster”. Caster works similar to the wheels on a shopping cart.

Caster is used to stabilize the steering of the car. If cars had no caster, it would be almost impossible to keep the car moving straight, especially at high speeds. Caster tries its best to ensure the steering stays straight.

However, too much caster makes the steering very heavy. Commercial cars have power steering but not ours. Therefore, we had to balance our steering weight or steering torque and our caster to get the optimal results.

DIY BATTERY

Due to the tight Formula Student rules, we will need to design and manufacture our own battery in-house. Using our initial estimates and goals for the battery capacity and the motor voltage requirements, we were able to determine the required battery capacity and voltage.

To decide the cell configuration within the battery, we also needed to select our cells. After thorough market research on lithium-ion cells, we found cells that were both affordable and reliable with decent power characteristics.

Using this we were able to calculate the number of cells we would need connected in series and parallel.

Hence, we were able to start the hardware design. This involved optimizing the electrical and mechanical integrity of the system while being rule compliant.

DESIGNING A CHASSIS

The chassis is a giant bracket that holds all the components and systems of the car together while keeping the driver and electronics safe and reducing any potential causes of error or failure.

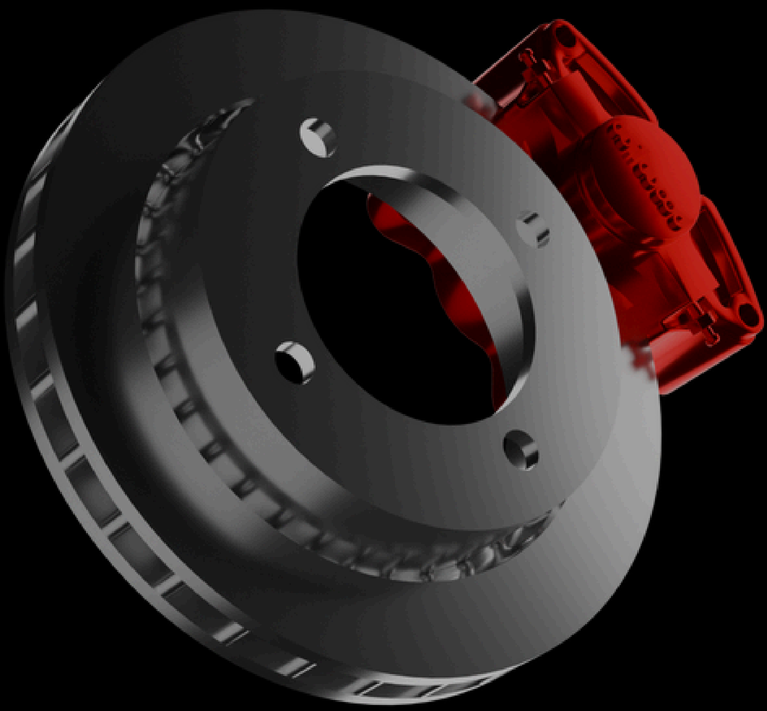
Although the chassis is the first component that needs to be designed for integration of all the systems, it is the last design that will be finalized.

The Formula Student rules are extensive for safety reasons. Therefore, the first step of the design process was to create all the basic parts of the chassis like the roll hoops, impact structures etc.

These were then integrated with the suspension pickup points and our Percy placement. After triangulating the nodes we were able to develop the first sketch of the chassis.

The design constantly changes as we add new components and do more static and dynamic load testing on the chassis.

MAKE IT STOP



The mechanical brakes of the car are extremely important for increasing our lap times. Being able to decelerate the car as quick as possible is crucial for this.

Simulations we were able to estimate the maximum deceleration speed possible before the tires lose grip. Using this we were able to calculate the forces required to bring the car to a stop as fast as possible.

Using this we were able to find the ideal rotor dimensions, calipers and pads that would not have a negative impact on our performance over a long period of time.

Brake biasing is the ratio between the front and rear braking force. With a rearward center of gravity, the rear brakes need to apply more force in order to lock both wheels at the same time. It also influences the dynamics of the car

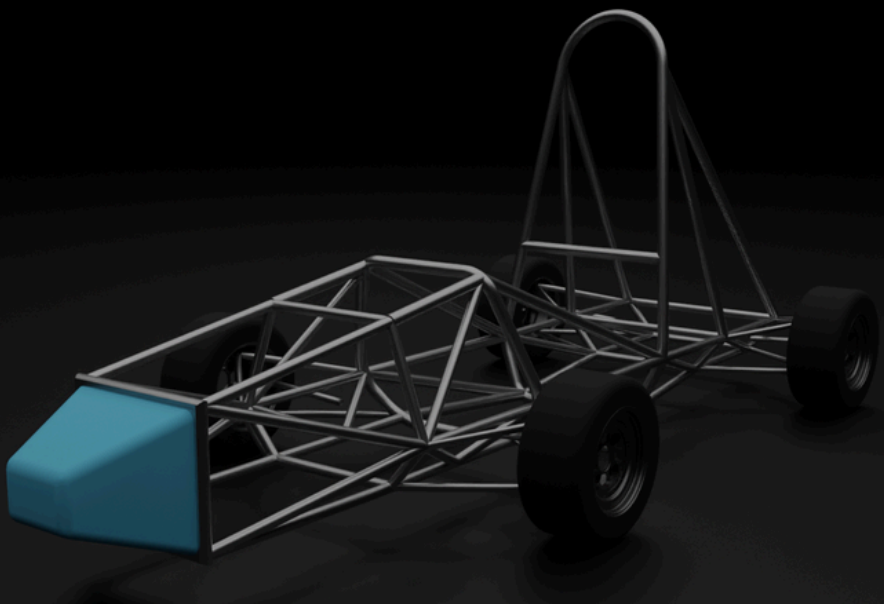
HOW YOU CAN SPONSOR US

What if you had the front row seat to engineering pioneers at work?

What if you were the reason these budding innovators were changing the future?

What if your institution could get their message to the rest of the world through Formula Student?

If your answer to any of these questions is yes, click here and check out what we have in store for our generous sponsors and learn how to become one yourself!



AUS

American University of Sharjah

AUTOMECH GROUP